Contribution Reflection

Reflection word count: 744

Contribution-focused paper

[1] Jacob O. Wobbrock and Julie A. Kientz. 2016. Research Contributions in Human-Computer Interaction. *Interactions* 23(3), 38-44. https://doi.org/10.1145/2907069

Research paper

[2] Faria Huq, Zora Zhiruo Wang, Frank F. Xu, Tianyue Ou, Shuyan Zhou, Jeffrey P. Bigham, and Graham Neubig. 2025. CowPilot: A Framework for Autonomous and Human-Agent Collaborative Web Navigation. In *Proceedings of NAACL 2025 (System Demonstrations)*, 163-172. https://aclanthology.org/2025.naacl-demo.17/

Reflection

Concepts from W&K

Wobbrock and Kientz (W&K) distinguish contributions by the knowledge they produce, and judge each type differently [1]. Artifact contributions arise from invention; systems are judged by "what they make possible and how they do so." Empirical contributions are findings from observation, judged on importance and soundness. Methodological contributions improve how research or practice is conducted, judged on utility, reproducibility, reliability, and validity. Theoretical contributions offer concepts and models, judged on novelty, soundness, and power to describe, predict, and explain. Dataset contributions supply a useful, representative corpus.

CowPilot

CowPilot is a Chrome extension in which an agent proposes each web action while a person can pause, take over, and resume [2]. On 27 WebArena tasks, with three authors as operators, collaborative mode with GPT-4o reached 95% accuracy with humans performing 15.2% of steps. Through this framework, I see one clear artifact contribution surrounded by contributions whose type is less settled.

Insight 1: The artifact is the arrangement, not the accuracy

The artifact contribution resides in the arrangement of collaboration, not in CowPilot's performance numbers. A proposed action runs after five seconds unless the user objects, and human actions enter the history available to the agent [2, sec. 2.1]. The design knowledge lies in making control transferable without treating intervention as the end of automation. W&K's system standard clarifies why I would not locate the contribution in the 95%: the same design reaches 0.81 with a smaller Llama model, below the 0.89 human-only baseline [2, sec. 4]. Handoff remains possible, but agent-driven completion falls from 0.52 to 0.05. This extends W&K's standard: for artifacts built on a swappable model, what they make possible must be stated relative to that model.

Insight 2: Demonstration or empirical contribution?

The evaluation is ambiguous between demonstrating the artifact and contributing empirical knowledge. The section is titled "Exemplar Findings" and exists to demonstrate usage, yet the abstract leads with its numbers and the introduction poses empirical questions, including what makes collaborations succeed. The answer is one qualitative paragraph [2, sec. 4.2]. As a demonstration, the evaluation only needs to show that collaboration works. As an empirical contribution, it would face W&K's soundness test, which three author-operators and an acknowledged ordering bias [2, sec. 6] cannot easily pass. I therefore read the numbers as artifact support and the observation that people intervene after repeated invalid actions as an exploratory lead. The gap may reflect a system demonstration with a larger study still underway: empirical ambitions precede the empirical contribution.

Insight 3: The metrics as a candidate method

CowPilot's metrics sit at the boundary between a system feature and a methodological contribution. The authors propose them as part of the framework, but distinguishing interventions from human steps, since one takeover can contain several steps [2, sec. 2.2], could be reused outside this interface. Under W&K's criteria, utility is plausible, but reproducibility, reliability, and validity are untested after one evaluation. For example, agent-driven completion only requires the agent to take the final step, so its validity is open. I would call this a candidate method, currently functioning as evaluation support.

Insight 4: Theory-like and dataset-like, but really artifact support

Two things that resemble other contribution types turn out to be artifact capabilities or support. CowPilot represents human and model as policies generating actions in a shared environment [2, sec. 2.1], which resembles a theoretical contribution. It can describe a trajectory, but it does not predict or explain who acts next; the account of when people intervene comes from observation instead. It is descriptive scaffolding, which may be why the authors never claim it as theory. Similarly, the paper promotes data collection [2, sec. 3.2] but describes no corpus: a capacity to create a dataset, not a dataset contribution.

How the contributions combine

These contributions are not parallel; they depend on one another. The artifact computes the metrics, the metrics define success in the evaluation, and the evaluation vouches for the artifact: "minimal" human intervention is judged by a step count CowPilot itself defines [2, sec. 2.2, 4.2]. Only the WebArena tasks and human-only baseline sit partly outside that loop, so the combination is coherent but largely self-supporting. This also extends W&K, whose CHI data treat multiple contributions as labels authors can check; 30.4% of submissions checked more than one. CowPilot suggests the relations between types matter as much as the labels. Presenting a combination effectively means stating which contribution is the claim, which are support, and what each can establish independently.

AI Use Disclosure

I used AI to help rephrase some passages, translate some text, and check grammar.
I completed the readings independently, developed the core ideas and analytical framework, and wrote the reflection myself.